

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PHY 222 — Past Questions & Materials
by Renance

Course: PHY 222
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 200, 300

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Paper 1 of 3 — PHY 222 - 2022/2023 - Second Semester - Physics II

Level 200 • Second Semester • PastQuestion

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Federal University of Petr le 1m Resources 'Esturun, Nigeria.

College of Scienc ,) epartment of Physics

'Second Se: e: er Examination

Academic Session: 2015/2016

Course Title: Electric Circuit & Networ!

Course Code: PAY 222 @ units)

'Time Allowed: 2hrs.30mins.

Analysis

Date: 30/08/2016

cot! ¢ Three Questions. Complete and ordered answers are

'Answer Question 1 and Any

rv ual meanings.

INSTRUCTIONS:

Note that all symbols have t!

preferred.to fragments.

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ta network to Wye network.

Question'

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(i) Derive the conversion formulas fo. (ran forming Wye network fo Delta network,
be balanced. 'Then = 4

" B.A Slate the © ind \ networks are said to

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iy Give one application of the Wye at 1D. & network.

alent resistance of 16 T stwork shown below at the terminal A-B. Take

C. Find the equiv

Rak =R~ReRe=RA5A. ne

'ondition for'which the Y

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PG ar ike terminals A-B Gin Figure 1) we co: cecied to a 20V source, what is the current shew 3 to the eircuit? 7

Gi) Obtain the power dissipated as) -at int eizeuit. ure urtaee a
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Figure 2 ya \e Figure 3 ove k

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7 Sind the current jy in the circuit below.

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<<“ (iii) Supenposition principle,

«B. () Explain the three (3) steps involv

Y In the circuit below, use superpos

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ed in perposition principle.

sition eorem to find V. »

20 80. ie

es) ae

Figure S(a)

C. For the circuit in Figure 5(b),

Ww

Question 5

- A. Explain (i) Thevenin's theorem,
 B. (i) Find the Thevenin's equivalent

Figure 6,

C. Find

$P_{R1} = 4 \text{ a. A !}$

Gi) Obtain the Currents through R,
 and the Norton equivalent circuit of

I_1 , we S (by

Find j when $V_1 = 12 \text{ v}$ and $V_2 = 24 \text{ v}$

(6 marks)

Gi) Norton theorem,

the circuit to the left of the

terminals a-b, for the circuit shown in

Figure 6, 16: and 36- ohms.

the circuit

it in Figure 7. - : why

32 v and R_{ab}

$T_s = 30 \text{ ms}$ and $f = 1 \text{ kHz}$

Figure 6 Figure 7 and Figure 8

(6 marks)

Question 6 features

Using nodal analysis, find the node voltages v_1 , v_2 , and v_3 in the circuit below.

Figure 3a 7 ;

20 v and 10 v (6 marks)

es)

Figure 8 and

Figure 5 and

Figure 4 - 0

Figure 6 and

6 marks

Figure 9 = 10 marks)

Figure 10 and

Figure 11 and

4 marks

Paper 2 of 3 — PHY 222 - 2022/2023 - Second Semester - Physics II

Level 200 • Second Semester • PastQuestion

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Federal University of Petroleum Resources Effurun, Nigeria.

College of Science, Department of Physics.

Second Semester Examination

Academic Session: 2016/2017

Course Title: Electric Circuit & Network Analysis

Course Code: PHY 222 (3 units)

Time Allowed: 3hrs. Date: 21/08/2017

INSTRUCTIONS: Answer Question 1 and Any other Three Questions. Complete and ordered answers are

preferred to fragments. Note that all symbols have their usual meanings.

Question 1

A. (i) Derive the conversion formulas for transforming Delta network to Wye network.

(ii) Derive the conversion formulas for transforming Wye network to Delta network.

B. (i) State the condition for which the Y and A networks are said to be balanced. Then write down the conversion formula under this condition.

(ii) Give one application of the Wye and Delta network.

C. Find the equivalent resistance of the network' shown below at the terminal A-B, Take

$R=R=R=R=R=$

(22marks)

“Question 2

A. (i) If the terminals A-B (in Figure 1) are connected to a 20V source, what is the current delivered to the circuit?

(ii) Obtain the power dissipated as heat in the circuit.

B. Use Kirchhoff's current law to find i_4 , i_1 , i_2 , and i_3 in Figure 2.

Where, $R_1=R_2=R_3=35\Omega$, $R_4=R_5=250$.

C. Use Kirchhoff's voltage law to obtain in Figure 3. 'a

1

i_1 , R_1

$I_1=2A$

ba |, A He [ye Ye

as, Be WW? 100) sa []so [hoa

Figure 2 Figure 3

(16marks)

Question 3

Using mesh analysis, find the node voltages i , in the circuit below.

62°

' t é 8.

Figure 4 (16marks)

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Question 4

A. In a linear circuit, explain the following (i) Homogeneity (ii) Additivity property and (iii) Superposition principle.

B. (i) Explain the three (3) steps involved in superposition principle.

(ii) In the circuit below, use superposition theorem to find V .

Figure 5(a) "Figure 5(b)

C. For the circuit in Figure 5(b), Find i when $V_1 = 12V$ and $V_2 = 24V$.

(6marks)

Question 5

A. Explain (i) Thevenin's theorem, (ii) Norton's theorem.

B. (i) Find the Thevenin's equivalent circuit to the left of the terminals a-b, for the circuit shown in Figure 6.

(ii) Obtain the currents through $R_1 = 6, 16, 26$ and 36 - ohms.

C. Find the Norton equivalent circuit of the circuit in Figure 7.

49 19 80

4 = S_a

Q_a

32V @ 0.1Ω on P_e , a_e

2V

b

8Q

Figure 6 Figure 7

(16marks)

Question 6

A(i) What do you understand by Laplace transform?

(ii) Determine the Laplace transform of the following: (i) $f(t) = u(t)$ (ii) $f(t) = e^{-t}u(t)$ (iii)

$f(t) = \sin \omega t u(t)$ (iv) $f(t) = 10 \cos \omega t u(t)$

(iii) List 4 properties of the Laplace transform with their equations

B. Using nodal analysis, find the node voltages V_1, V_2, V_3 , and V_4 in the circuit below.

a

20V y ,

307

$V_3 V_a$

12

Figure 8

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Paper 3 of 3 — PHY 222 - 2022/2023 - Second Semester - Physics

Level 300 • Second Semester • PastQuestion

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Ans: When the input is increased by constant the output also increases.

$$V = 11$$

$$I = 0.1A$$

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College of Science, Department of Physics.
Second Semester Examination

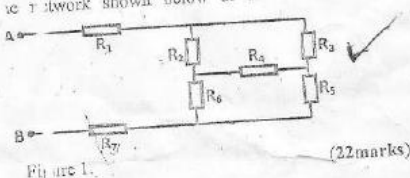
Academic Session: 2015/2016
Course Title: Electric Circuit & Network Analysis
Course Code: PHY 222 (3 units)
Time Allowed: 2hrs.30mins.

Date: 30/08/2016

INSTRUCTIONS: Answer Question 1 and Any other Three Questions. Complete and ordered answers are preferred to fragments. Note that all symbols have their usual meanings.

Question 1

- (i) Derive the conversion formulas for transforming Delta network to Wye network.
(ii) Derive the conversion formulas for transforming Wye network to Delta network.
B. (i) State the condition for which the Y and Δ networks are said to be balanced. Then write down the conversion formula under this condition.
(ii) Give one application of the Wye to Δ network.
C. Find the equivalent resistance of the network shown below at the terminal A-B. Take $R_1 = R_2 = R_3 = R_4 = R_5 = R_6 = 5\Omega$.



WEM
Question 2

- (i) If the terminals A-B (in Figure 1) are connected to a 20V source, what is the current delivered to the circuit?
(ii) Obtain the power dissipated as heat in the circuit.
B. Use Kirchhoff's current law to find i_1, i_2, i_3, i_4 , and i_5 in Figure 2.
Where, $R_1 = R_2 = R_3 = 35\Omega$, $R_4 = R_5 = 25\Omega$.
C. Use Kirchhoff's voltage law to obtain i_6 in Figure 3.

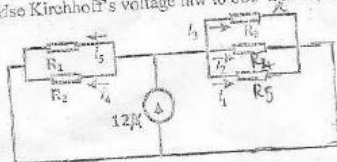


Figure 2

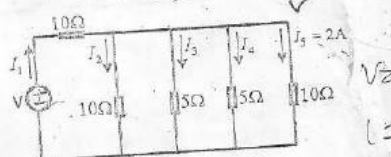


Figure 3

Question 3

Using mesh analysis, find the current i_0 in the circuit below.

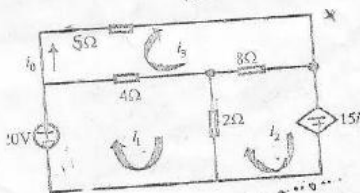


Figure 4

Additional

Or circuit the sum of the output
voltage to the total voltage of the output.
 $V = IR$

mob (17+12) gives
154 - 1563

Question 4

- A. In a linear circuit, explain the following (i) Homogeneity (ii) Additivity property and (iii) Superposition principle.
 B. (i) Explain the three (3) steps involved in superposition principle.
 (ii) In the circuit below, use superposition theorem to find V .

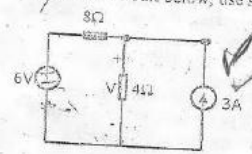


Figure 5(a)

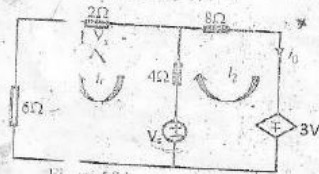


Figure 5(b)

- C. For the circuit in Figure 5(b), Find I_0 when $V_1 = 12V$ and $V_2 = 24V$.

(16marks)

Question 5

- A. Explain (i) Thevenin's theorem, (ii) Norton theorem.
 B. (i) Find the Thevenin's equivalent circuit to the left of the terminals a-b, for the circuit shown in Figure 6.
 (ii) Obtain the currents through $R_1 = 6, 16, 36$ ohms.
 C. Find the Norton equivalent circuit of the circuit in Figure 7.

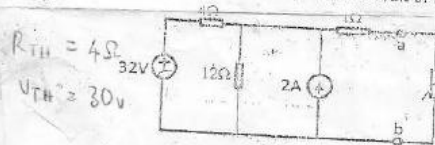


Figure 6

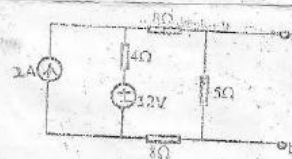


Figure 7

CH 4 E16

CH 4 E22

(16marks)

Question 6

Using node analysis, find the node voltages V_1, V_2, V_3 and V_4 in the circuit below.

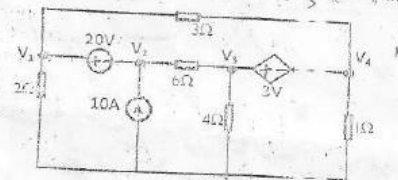


Figure 8

Using KVL

$$-6 + 8I_2 + 4I_2 = 0$$

$$-6 + 12I_2 = 0$$

$$-6 = -12I_2$$

$$I_2 = -\frac{6}{-12} = 0.5A$$

$$V_2 = I_2 R_2 = 4 \times 0.5 = 2V$$

$$I = V_2 / R_2 = 2 / 4 = 0.5A$$

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(16marks)